ADS-505 Team Project Proposal Form

Work with your teammates to complete the three sections of this form. Designate one team member to submit your completed form.

Project Team

Team Number (Canvas Project Group #):

Team Members (Full Names):

1. Amir Badeanlo
2.
3.

Team GitHub link (required)**:** https://github.com/amirarsalan4543/ADS_505

Have you coordinated who will submit the deliverables (e.g., a team leader, delegate, reprentative, etc.)? solo project

How many times have your team members met so far? solo project

List the specific contributions that each team member is providing for the Final Team Project in the table below. All students on the team should contribute equally to the Final Team Project (refer to the Team Project requirements in Canvas).

Team Member 1 (Name)	Team Member 2 (Name)	Team Member 3 (if applicable) (Name)
solo project	solo project	solo project

Comments/ Roadblocks: solo project

Project Summary

Project title: Mitigating Retail Trade-In Liquidity Risk: Modeling Pokémon TCG Secondary Market Price Spreads

Project short description (objectives): Analyze secondary market card attributes and build predictive statistical models that quantify the price spread between TCGplayer theoretical market prices and low floor liquidation prices. The objective is to provide local game stores (LGS) with a data-driven framework to dynamically adjust trade-in buy margins, reducing dead-stock risk while maintaining transparent, fair cash-out pricing for consumers.

Dataset (name, source, link, citation, etc.):
Name: Pokemon-TCG-Data
Source / Curator: PokemonTCG.io
Link: https://github.com/PokemonTCG/pokemon-tcg-data

Dataset description (source, number of variables, size of the dataset, etc.):
contains over 15,000 card records spanning all major English releases from the 1999 Base Set to modern expansions.

Card Metadata: Card ID, name, set ID, series, set release date, artist, supertype (Pokémon, Trainer, Energy), subtypes, and Pokédex number.

Rarity & Finishes: Official rarity designation (Common, Uncommon, Rare Holo, Secret Rare, etc.) and finish types (Normal, Holofoil, Reverse Holofoil).

Market Metrics (TCGplayer): Multi-tier pricing in USD per finish variant, including low price, mid price, high price, and market price.

Project Proposal

In one or two pages, describe the project your team will pursue using the provided outline.

Purpose:
To develop a model that predicts price spreads and liquidity risk in secondary trading card markets – Specifically, the Pokemon Trading Card Game (TCG). Local game stores (LGS) operate on thin margins and face financial vulnerability when acquiring inventory through trade-ins. By identifying which card features correlate with high divergence between theoretical market valuations and actual floor liquidation prices, this project provides retail managers with an objective decision-support tool to replace rigid, blanket trade-in formulas with dynamic risk-adjusted margins.

Background:
The collectible trading card secondary market has experienced exponential growth, evolving from casual hobby trading into a volatile alternative asset class. Centralized platforms like TCGplayer publish aggregate benchmark statistics -- most notably the "Market Price," which is calculated as an average of recent completed transactions. Because retail stores do not have the time to individually analyze market depth for hundreds of incoming cards during counter transactions, the industry standard has defaulted to a flat-rate discount heuristic (e.g., paying a flat 50% in cash or 70% in store credit against published market prices). While this flat model functions adequately for highly liquid cards, it breaks down significantly when applied indiscriminately across thousands of collectible variants.

Current Situation:
In practice, a card's theoretical "Market Price" regularly decouples from its "Low Price". "Low Price" is the immediate clearing price required to liquidate inventory quickly. For example, a card may show a historical market value of $20 based on occasional sales, but carry an active floor listing of $11 due to low organic demand, overprinting, or declining tournament play. Under a standard 50% cash trade-in policy, a retailer pays $10 for a card worth $11 on the immediate floor. After accounting for overhead, platform processing fees, and potential price drops during holding periods, the store incurs a net loss. Retail clerks currently lack an empirical method to evaluate trade-in submissions dynamically, leaving the store to absorb downside.

Expected Conclusion:
This project expects to identify the primary card attributes that demonstrat statistically significant correlations with wide divergence between theoretical market prices and low floor liquidation prices. By quantifying these underlying predictors, the analysis will yield a predictive statistical framework that evaluates downside spread and liquidity risk for incoming trade-in items.

This framework will equip retail counter staff to transition away from rigid, blanket buy rates and dynamically tailor payout percentages based on empirical risk, allowing stores to prevent illiquid dead stock and offer transparent, equitable valuations to players.